Quantum Probability Simulator

Day 20 | 30-Day Quantum Computing Challenge — Mini Project

Quick Recap. In Day 19, a Hadamard gate prepared a qubit with equal probabilities for measuring 0 or 1. Day 20 studies why observed percentages fluctuate and how the typical fluctuation decreases as more shots are collected.

1. Objective

Repeat the ideal quantum coin-flip experiment for **10, 100, 1,000 and 10,000 shots**. Compare the measured fractions against the theoretical 50% and visualize how a running estimate evolves.

2. Circuit and Quantum Principle

```
|0> ---- [ H ] ---- [ Measure ] ---- c
```

The Hadamard gate prepares **|+> = (|0> + |1>) / sqrt(2)**. By the Born rule, **P(0) = P(1) = 0.5**. A shot is one preparation-and-measurement trial.

3. Two Ways to Compare Measurements

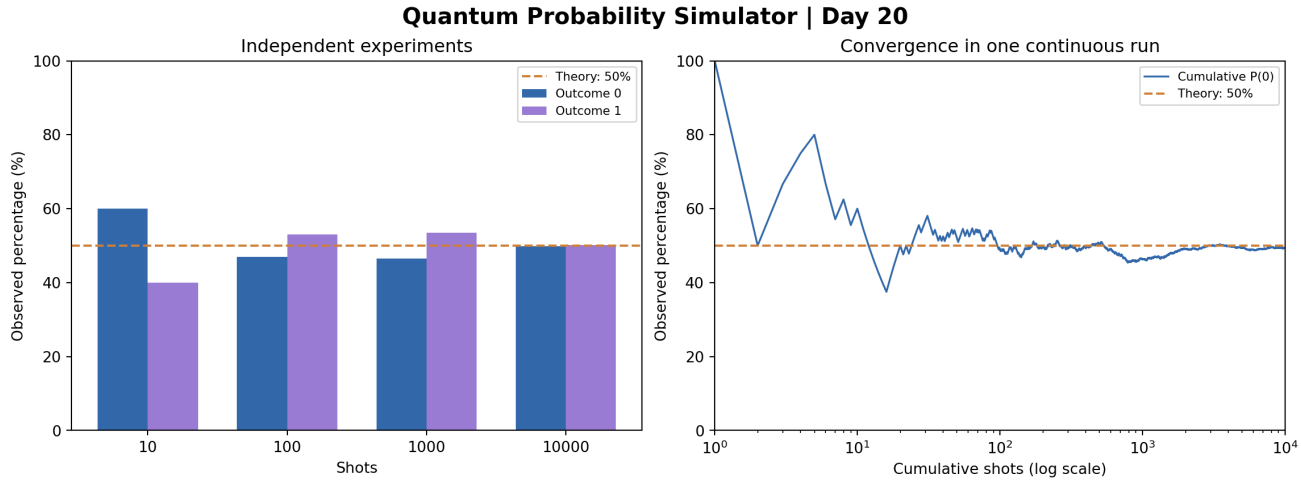
Independent runs: use a new batch of measurements at each shot count and compare the resulting bar chart. **Cumulative run:** collect one 10,000-shot sequence and track the fraction of zeros after each shot. This gives a clearer view of statistical convergence.

4. Example Simulation Results

Shots	Measured 0	Measured 1	P(0)	P(1)
10	6	4	60.0%	40.0%
100	47	53	47.0%	53.0%
1,000	465	535	46.5%	53.5%
10,000	4989	5011	49.9%	50.1%

Seeded NumPy sampling of the ideal 50/50 measurement distribution; values will differ with another seed.

5. Visual Results



Left: independent experiments. **Right:** the running proportion of zeros from one continuous experiment. The dashed line marks the exact theoretical 50%.

6. Why Does It Converge?

If the shots are independent and equally likely, the number of zeros follows a binomial distribution. The standard deviation of the observed fraction is **sqrt[p(1-p)/n] = 1/(2 sqrt(n))** when p = 0.5. Typical percentage fluctuations therefore decrease approximately as **1/sqrt(n)**. A larger sample is not guaranteed to be closer on every individual run.

Typical one-standard-deviation sampling fluctuation

Shots	Approx. standard deviation of observed P(0)
10	15.81 percentage points
100	5.00 percentage points
1,000	1.58 percentage points
10,000	0.50 percentage points

7. How to Execute the Experiment

1. Open `quantum_probability_simulator.ipynb` in Jupyter or Google Colab.
2. Run the NumPy sampling cells to generate independent shot-count results.
3. Run the visualization cells to display and save `results.png`.
4. Inspect the standard-deviation comparison.
5. Optionally install `qiskit` and `qiskit-aer` and execute the corresponding H-and-measure circuit on an ideal circuit simulator.

8. What the Notebook Contains

A reproducible seeded sampling experiment, a comparison bar chart, a cumulative convergence curve, a simple statistical explanation and an optional Qiskit Aer circuit. The default NumPy path runs without quantum-specific packages.

9. Interpretation and Limitations

Theoretical probabilities are 50% for both outcomes; measured percentages vary. A finite-sample graph can oscillate above and below 50%, and a bigger sample does not always improve the estimate in a single comparison. These results simulate ideal quantum measurement probabilities and do **not** demonstrate quantum hardware performance or certified physical randomness.

10. Common Mistakes

• Assuming 10,000 shots must produce exactly 5,000 zeros.
• Claiming that every increase in shots guarantees a closer result.
• Confusing separate shot-count experiments with a cumulative curve.
• Describing pseudorandom classical simulation as hardware-generated quantum randomness.

Summary: What to Remember

A Hadamard-prepared qubit has theoretical P(0) = P(1) = 0.5. Repeated measurements approximate those probabilities; typical error shrinks with the square root of the number of shots. **Next: Day 21 — Quantum Random Number Generator.**